

From four levels to the Dicke model

removing the intermediate state

September 14, 2026

Abstract

A companion note carries the four-level cavity Rydberg-EIT problem all the way to a Dicke model. This one stops earlier, at the three-level cavity Hamiltonian $H^{(3)}$ of Eq. (25), in order to treat Step 3 — the removal of the intermediate state $|2\rangle$ — in full. That step is where the Raman coupling $\lambda = g_0\Omega_c/2\Delta$ comes from, along with the light shifts that accompany it and the spontaneous emission it inherits, and it is the only place in the reduction where anything is given up. We derive it from the eigenvalue problem, assuming no prior acquaintance with perturbation theory: eliminating one component of an eigenvector is exact linear algebra, and a single approximation — neglecting the eigenvalue beside the detuning — turns the exact result into the familiar second-order formula. The same calculation yields the decay rates the surviving states inherit, the invariant $\lambda^2/\gamma_3 = g_0^2/\Gamma_e$, and the error of the reduction, which proves to be the residual population of the state removed. Throughout $\hbar = 1$.

1 The exact Hamiltonian

Four atomic levels with laboratory energies $\omega_1 = 0 < \omega_2 < \omega_3 < \omega_4$:

$$|1\rangle = |5S_{1/2}\rangle, \quad |2\rangle = |5P_{3/2}\rangle, \quad |3\rangle = |53D_{5/2}\rangle, \quad |4\rangle = |54P_{3/2}\rangle,$$

numerically $\omega_2/2\pi = 384.25$, $\omega_3/2\pi = 1008.82$ THz, and $(\omega_4 - \omega_3)/2\pi \simeq 10$ GHz. Writing $\sigma_{jk} = |j\rangle\langle k|$, and with no approximations whatsoever,

$$\begin{aligned} H = & \underbrace{\sum_{j=1}^4 \omega_j \sigma_{jj} + \omega_{\text{cav}} a^\dagger a}_{\text{free}} + \underbrace{g_0 (a + a^\dagger)(\sigma_{12} + \sigma_{21})}_{\text{cavity, } |1\rangle \leftrightarrow |2\rangle} \\ & + \underbrace{\Omega_c \cos(\nu_c t) (\sigma_{23} + \sigma_{32})}_{\text{control laser}} + \underbrace{\Omega_\mu \cos(\nu_\mu t) (\sigma_{34} + \sigma_{43})}_{\text{microwave}}. \end{aligned} \quad (1)$$

The classical fields carry their full time dependence; the cavity coupling carries both a and $a^\dagger$. Nothing has been dropped.

2 Step 1: the rotating frame

Transform with $\tilde{H} = U H U^\dagger + i\dot{U}U^\dagger$, choosing

$$U(t) = \exp\left\{it[\omega_L a^\dagger a + \omega_L \sigma_{22} + (\omega_L + \nu_c) \sigma_{33} + (\omega_L + \nu_c + \nu_\mu) \sigma_{44}]\right\}, \quad (2)$$

i.e. each level is rotated at the *sum of the drive frequencies needed to reach it* (ω_L is a reference near the cavity resonance). This is an exact change of picture. Its whole effect is on the diagonal:

level	rotated at	lab energy	rotating-frame energy
$ 1\rangle$	0	0	0
$ 2\rangle$	ω_L	$2\pi \times 384$ THz	$\Delta \equiv \omega_2 - \omega_L$ ($\sim$ GHz)
$ 3\rangle$	$\omega_L + \nu_c$	$2\pi \times 1009$ THz	$\delta \equiv \omega_3 - \omega_L - \nu_c$ ($\sim$ MHz)
$ 4\rangle$	$\omega_L + \nu_c + \nu_\mu$	$\omega_3 + 2\pi \times 10$ GHz	$\delta + \Delta_\mu$, $\Delta_\mu \equiv (\omega_4 - \omega_3) - \nu_\mu$

This answers the obvious objection. $|3\rangle$ and $|4\rangle$ are *not* degenerate — they are 10 GHz apart. But the frame rotates $|4\rangle$ faster than $|3\rangle$ by exactly ν_μ , and ν_μ is chosen close to that splitting. The 10 GHz is subtracted off; what remains on the diagonal is only the residual *microwave detuning* Δ_μ , of order MHz. Setting $\Delta_\mu = 0$ makes the two entries equal — and that is a statement about the *drive*, not about the atom.

3 Step 2: the rotating-wave approximation

In the new frame the cavity term $a^\dagger \sigma_{21}$ picks up $e^{2i\omega_L t}$, and $\cos(\nu t) \sigma_{jk}$ splits into a static piece and one at $e^{2i\nu t}$. Dropping the fast pieces requires

$$g_0, \Omega_c, \Omega_\mu \ll \omega_L, \nu_c, \nu_\mu, \quad (\text{A1})$$

which is spectacularly well satisfied (MHz against THz, and MHz against the 10 GHz microwave). The result is time-independent and still exact to this order:

$$\tilde{H} = \omega_c a^\dagger a + \begin{pmatrix} 0 & g_0 a^\dagger & 0 & 0 \\ g_0 a & \Delta & \Omega_c/2 & 0 \\ 0 & \Omega_c/2 & \delta & \Omega_\mu/2 \\ 0 & 0 & \Omega_\mu/2 & \delta + \Delta_\mu \end{pmatrix}, \quad \omega_c \equiv \omega_{\text{cav}} - \omega_L. \quad (3)$$

Note the matrix is tridiagonal: each field connects only neighbouring levels.

4 Step 3: eliminating the intermediate state

The Hamiltonian (3) is a chain: the cavity talks to $|2\rangle$, $|2\rangle$ talks to $|3\rangle$, $|3\rangle$ talks to $|4\rangle$. Of these, $|2\rangle$ is the odd one out. It is the only short-lived state in the problem, radiating at $\Gamma_e/2\pi \simeq 6$ MHz, and it is detuned by $\Delta/2\pi \sim 100$ MHz, far more than any other energy in the rotating frame. A state that far off resonance is barely populated, and one might hope to do without it.

One cannot simply delete it. It is the only bridge between the cavity photon and the Rydberg state: set $|2\rangle$ aside and $|1\rangle$ and $|3\rangle$ stop talking altogether. The object of this section is therefore to remove $|2\rangle$ as a degree of freedom while *keeping* the connection it provides, and to do so in a controlled way, so that we know what has been given up.

4.1 Reducing to a finite matrix

The entries of Eq. (3) contain the cavity operators a and $a^\dagger$, which act on an infinite space. One observation makes the problem finite. Every term in Eq. (3) either leaves the atom alone, or moves it up one level while destroying a photon ($a \sigma_{21}$), or moves it down one while creating one ($a^\dagger \sigma_{12}$), or moves it between $|2\rangle, |3\rangle, |4\rangle$ with no photon involved at all. The quantity

$$n = a^\dagger a + \sigma_{22} + \sigma_{33} + \sigma_{44} \quad (\text{photons} + \text{atomic excitation}) \quad (4)$$

is therefore conserved: $[\tilde{H}, n] = 0$. The Hamiltonian is block diagonal in n , and we may work in one block at a time and lose nothing.

The block with n excitations is spanned by four states, since the atom can hold at most one:

$$|1, n\rangle, \quad |2, n-1\rangle, \quad |3, n-1\rangle, \quad |4, n-1\rangle, \quad (5)$$

where the second label is the photon number. Their matrix elements follow from $a^\dagger|n-1\rangle = \sqrt{n}|n\rangle$; every diagonal entry carries a common $\omega_c(n-1)$, which we subtract because a constant shifts all eigenvalues equally. What is left is an ordinary 4×4 matrix of numbers,

$$H^{(n)} = \begin{pmatrix} \omega_c & g & 0 & 0 \\ g & \Delta & \Omega_c/2 & 0 \\ 0 & \Omega_c/2 & \delta & \Omega_\mu/2 \\ 0 & 0 & \Omega_\mu/2 & \delta + \Delta_\mu \end{pmatrix}, \quad g \equiv g_0\sqrt{n}. \quad (6)$$

The $\sqrt{n}$ is stimulated emission: a cavity holding n photons couples $\sqrt{n}$ times more strongly than one holding a single photon. (Do not confuse this n , the photon number, with the atom number N , which does not appear until §4.5.)

4.2 Eliminating one component exactly

We want the eigenvalues of Eq. (6). Write the eigenvector as $c = (c_1, c_2, c_3, c_4)$ and the eigenvalue equation $H^{(n)}c = Ec$ out row by row. It is useful to give a name to the second column, the couplings that reach $|2\rangle$:

$$V_1 = g, \quad V_3 = \Omega_c/2, \quad V_4 = 0. \quad (7)$$

That $V_4 = 0$ is a statement about the apparatus: the microwave connects $|3\rangle$ to $|4\rangle$ and never touches $|2\rangle$. With this notation the four rows read

$$E c_m = \sum_{k \neq 2} H_{mk} c_k + V_m c_2, \quad m = 1, 3, 4, \quad (8)$$

$$E c_2 = \Delta c_2 + \sum_{k \neq 2} V_k c_k. \quad (9)$$

Equation (9) is a single linear equation for the single unknown c_2 , and we may simply solve it:

$$c_2 = \frac{1}{E - \Delta} \sum_{k \neq 2} V_k c_k \quad (10)$$

This says that $|2\rangle$ is not an independent actor: whatever amplitude it carries is dictated by the states that drive it, suppressed by the energy denominator $E - \Delta$. Substituting Eq. (10) back into Eq. (8) removes c_2 from the problem entirely:

$$E c_m = \sum_{k \neq 2} \left[\underbrace{H_{mk}}_{\text{direct}} + \underbrace{\frac{V_m V_k}{E - \Delta}}_{\text{via } |2\rangle} \right] c_k, \quad m, k \in \{1, 3, 4\}. \quad (11)$$

Nothing has been approximated. Equation (11) is a 3×3 problem with exactly the same eigenvalues as the 4×4 we started from.¹ The bridge survives elimination: even though $H_{13} = 0$, the second term connects $|1\rangle$ and $|3\rangle$ through the product $V_1 V_3$.

The price is that E now appears on both sides. Equation (11) is not an ordinary eigenvalue problem but an implicit one, to be solved self-consistently.

¹Three of them, at least. The fourth eigenvalue, the one near Δ that is mostly $|2\rangle$, is also a solution of Eq. (11); it is not lost, merely of no interest here.

4.3 The one approximation

Look at the size of the energy denominator. The eigenvalues E we care about are those of the slow states: they lie within a few MHz of zero, set by ω_c , δ and Ω_μ . The detuning Δ is a hundred times larger. To the extent that $|E| \ll |\Delta|$ we may replace $E - \Delta$ by $-\Delta$, and the implicit problem collapses to an ordinary matrix:

$$\boxed{H_{mk}^{\text{eff}} = H_{mk} - \frac{V_m V_k}{\Delta}} \quad m, k \in \{1, 3, 4\}. \quad (12)$$

This is the whole of Step 3. It is what is usually called *second-order perturbation theory*, and the name describes the formula: each correction is a product of *two* coupling matrix elements — second order in the fields — divided by an energy denominator. Read from right to left, $V_m V_k / (-\Delta)$ is one term per *path* $m \rightarrow 2 \rightarrow k$: amplitude to hop up to $|2\rangle$, amplitude to hop back down to $|k\rangle$, penalised by how far off resonance the detour was.

For $m = k$ the formula reduces to something familiar from elementary quantum mechanics, the repulsion of two levels that are coupled: $|m\rangle$ is pushed by $-V_m^2/\Delta$, away from $|2\rangle$. Since $\Delta > 0$ here, $|2\rangle$ lies above and the states below it are pushed down. The new content of Eq. (12) is the case $m \neq k$, which is what one needs when the states being kept are close together in energy compared with the one being removed — exactly our situation, MHz against 100 MHz. It is sometimes called quasi-degenerate perturbation theory for that reason.

Two remarks before we use it. First, the approximation made is precisely the neglect of E beside Δ in one denominator; §4.7 turns that into a number. Second, this is not a new kind of approximation but the one already made in Step 2. There we discarded a field that oscillated fast compared with everything it drove, and kept its second-order residue; here we discard the independent motion of an amplitude that oscillates fast, at Δ , compared with everything that drives it, and keep its second-order residue. Appendix A carries out the same elimination in the time domain, where the parallel is explicit.

4.4 Evaluating the four paths

There are nine entries (m, k) in Eq. (12), but $V_4 = 0$ kills any path through $|4\rangle$, leaving four:

path	$V_m V_k$	correction $-V_m V_k / \Delta$	what it is
$1 \rightarrow 2 \rightarrow 1$	g^2	$-g^2 / \Delta$	a shift of $ 1\rangle$, one per photon
$3 \rightarrow 2 \rightarrow 3$	$\Omega_c^2 / 4$	$-\Omega_c^2 / 4\Delta$	a light shift of $ 3\rangle$
$1 \rightarrow 2 \rightarrow 3$	$g \Omega_c / 2$	$-g \Omega_c / 2\Delta$	the Raman coupling
$3 \rightarrow 2 \rightarrow 1$	$\Omega_c / 2 \cdot g$	$-g \Omega_c / 2\Delta$	the same, by symmetry

so that, in the basis $(|1, n\rangle, |3, n-1\rangle, |4, n-1\rangle)$,

$$H^{\text{eff}} = \begin{pmatrix} \omega_c - \frac{g^2}{\Delta} & -\lambda\sqrt{n} & 0 \\ -\lambda\sqrt{n} & \delta - \frac{\Omega_c^2}{4\Delta} & \Omega_\mu/2 \\ 0 & \Omega_\mu/2 & \delta + \Delta_\mu \end{pmatrix}, \quad \boxed{\lambda = \frac{g_0 \Omega_c}{2\Delta}} \quad (13)$$

where we have used $g = g_0 \sqrt{n}$ to pull the photon number out front.

Finally we undo the block decomposition. A matrix element proportional to $\sqrt{n}$ between $|1, n\rangle$ and $|3, n-1\rangle$ is what the operator $a^\dagger$ produces, and a diagonal entry proportional to n is

what $a^\dagger a$ produces. Reassembling the blocks into one operator,

$$\tilde{H}^{(3)} = \omega_c a^\dagger a + \begin{pmatrix} -\frac{g_0^2 a^\dagger a}{\Delta} & -\lambda a^\dagger & 0 \\ -\lambda a & \delta - \frac{\Omega_c^2}{4\Delta} & \Omega_\mu/2 \\ 0 & \Omega_\mu/2 & \delta + \Delta_\mu \end{pmatrix}. \quad (14)$$

The ordering $a^\dagger a$ rather than $a a^\dagger$ is not a convention but a count: the shift of $|1\rangle$ came out proportional to n , the number of photons available to be absorbed, and an empty cavity offers none. Had we written $a a^\dagger$ we would be claiming a shift $\propto n+1$, and a vacuum cavity would shift the ground state — which the exact matrix (6) plainly does not do, since $g = g_0\sqrt{n}$ vanishes at $n = 0$.

4.5 What the three new terms are

The Raman coupling, $-\lambda(a^\dagger\sigma_{13} + a\sigma_{31})$. A cavity photon is absorbed on $|1\rangle \rightarrow |2\rangle$ and a control photon is emitted on $|2\rangle \rightarrow |3\rangle$, with $|2\rangle$ visited only virtually. This is the term the whole scheme is built on. Note that it is first order in *each* field separately: halving the control power halves λ , and it is Ω_c , not Ω_c^2 , that sets the coupling. The minus sign is a phase convention — redefining $|3\rangle \rightarrow -|3\rangle$ removes it — and we drop it from here on.

The control light shift, $-\Omega_c^2/4\Delta$ on $|3\rangle$. A constant displacement of the Rydberg level, absorbed by redefining the two-photon detuning $\delta \rightarrow \delta - \Omega_c^2/4\Delta$. Harmless as bookkeeping; not harmless as physics, for the reason set out in the box at the end of this section.

The dispersive shift, $-(g_0^2/\Delta) a^\dagger a$ on $|1\rangle$. This one multiplies $a^\dagger a$, so it is not an atomic energy at all but a shift of the *cavity* frequency, conditioned on the atom being in $|1\rangle$. With N atoms, all of them in $|1\rangle$ in the normal state of affairs, it adds up:

$$\omega_c \longrightarrow \omega_c - \frac{N g_0^2}{\Delta}. \quad (15)$$

For $N = 10^6$, $g_0/2\pi = 1$ MHz and $\Delta/2\pi = 100$ MHz this is $2\pi \times 10$ GHz, against an $\omega_c/2\pi$ of order 1 MHz. It is not a small correction to be dropped but a wholesale redefinition of where the cavity resonance sits, dealt with in the laboratory by locking to the shifted line. What matters for a sensor is that it moves when N or Δ moves.

4.6 The cost: inherited spontaneous emission

So far $|2\rangle$ has been treated as a stable level. It is not: it decays at $\Gamma_e/2\pi \simeq 6$ MHz. The standard way to carry a decaying level through a Hamiltonian calculation is to give its energy a negative imaginary part, $\Delta \rightarrow \Delta - i\Gamma_e/2$, so that an amplitude sitting in it dies as $e^{-\Gamma_e t/2}$. Nothing in the derivation above cared whether Δ was real, so Eq. (12) still holds; the denominators are now complex. Expanding for $|\Delta| \gg \Gamma_e$,

$$\frac{1}{\Delta - i\Gamma_e/2} \simeq \frac{1}{\Delta} + \frac{i\Gamma_e}{2\Delta^2}, \quad (16)$$

and the two diagonal paths pick up imaginary parts, that is, decay rates:

$$\boxed{\gamma_3 = \frac{\Omega_c^2}{4\Delta^2} \Gamma_e \quad (\text{Rydberg state}), \quad \kappa_{\text{abs}} = \frac{g_0^2}{\Delta^2} \Gamma_e \quad (\text{cavity})} \quad (17)$$

Each is Γ_e multiplied by the residual population that the corresponding drive maintains in $|2\rangle$ — $(\Omega_c/2\Delta)^2$ from the control laser, $(g_0/\Delta)^2$ per photon from the cavity. The elimination has not abolished spontaneous emission; it has diluted it in proportion to how little $|2\rangle$ is occupied. At $\Omega_c/2\pi = 5$, $g_0/2\pi = 1$ and $\Delta/2\pi = 100$ MHz these come to $2\pi \times 3.75$ and 0.60 kHz, which agree with the exact complex eigenvalues to better than 0.5%.

At first sight this invites a free lunch. Both λ and γ_3 shrink as the detuning grows, but at different rates, $\lambda \propto \Delta^{-1}$ against $\gamma_3 \propto \Delta^{-2}$, so detuning further appears to cost coupling more slowly than it buys coherence. It does not, and the cleanest way to see this is to form the combination in which both free parameters cancel:

$$\boxed{\frac{\lambda^2}{\gamma_3} = \frac{g_0^2 \Omega_c^2 / 4\Delta^2}{\Omega_c^2 \Gamma_e / 4\Delta^2} = \frac{g_0^2}{\Gamma_e}} \quad (18)$$

Neither Δ nor Ω_c survives. Detuning further and raising the control power to restore λ simply slides the system along a line of constant λ^2/γ_3 ; Step 3 offers no leverage on it at all. Dividing by the cavity linewidth κ , and using that N atoms enhance the coupling to $\lambda\sqrt{N}$ while leaving the single-atom γ_3 alone, this invariant is the familiar cooperativity,

$$\frac{N\lambda^2}{\kappa\gamma_3} = \frac{Ng_0^2}{\kappa\Gamma_e} = NC. \quad (19)$$

Atom number, cavity finesse and the choice of atomic transition are what improve the instrument; Δ and Ω_c only decide where along the line it operates. Cavity absorption is the exception that proves the rule: $\lambda^2/\kappa_{\text{abs}} = \Omega_c^2/4\Gamma_e$ *does* reward control power, precisely because κ_{abs} contains no Ω_c .

4.7 How accurate, and when valid

The single approximation was to drop E beside Δ in Eq. (10). Restoring it,

$$\frac{1}{E - \Delta} = -\frac{1}{\Delta} \left(1 + \frac{E}{\Delta} + \dots \right), \quad (20)$$

the first neglected term in Eq. (11) is $-E V_m V_k / \Delta^2$. Its effect on an eigenvalue is its expectation value in the corresponding eigenvector, $-E(\sum_k V_k c_k)^2 / \Delta^2$, and comparing with Eq. (10) the sum is just $(E - \Delta)c_2$. The error therefore takes a form with no free parameters and a transparent meaning:

$$\boxed{E_{\text{exact}} - E_{\text{eff}} \simeq -E|c_2|^2} \quad (21)$$

The fractional error in any energy is the residual population of the state we eliminated. Two consequences follow. Since $|c_2|^2$ falls as Δ^{-2} while λ falls only as Δ^{-1} , the description becomes *relatively* more accurate as one detunes, even as the physics it describes grows weaker. And a state that contains no $|3\rangle$ — a photon with the atom in the ground state — is reproduced far more accurately than a dressed state, because only g_0 , and not the much larger Ω_c , connects it to $|2\rangle$.

Figure 1 checks both claims against exact diagonalisation of Eq. (6). Because the reduction to a 4×4 block was exact, this comparison involves no truncation anywhere.

Collecting the conditions under which the step is justified,

$$|\Delta| \gg \Omega_c, \quad g_0\sqrt{n}, \quad \Omega_\mu, \quad |\delta|, \quad |\omega_c|, \quad \Gamma_e. \quad (\text{A2})$$

The first two are the couplings V_m that reach $|2\rangle$; the middle three are the energies E of the states retained, which is what had to be small beside Δ ; the last is the width of the level removed. All six enter, and the list is worth keeping whole: a treatment that checks only $|\Delta| \gg \Omega_c, \Gamma_e$ will be caught out at large photon number, where $g_0\sqrt{n}$ creeps up on Δ . At $\Delta/2\pi = 100$ and $g_0/2\pi = 1$ MHz the elimination holds to $n \sim 10^3$ and no further.

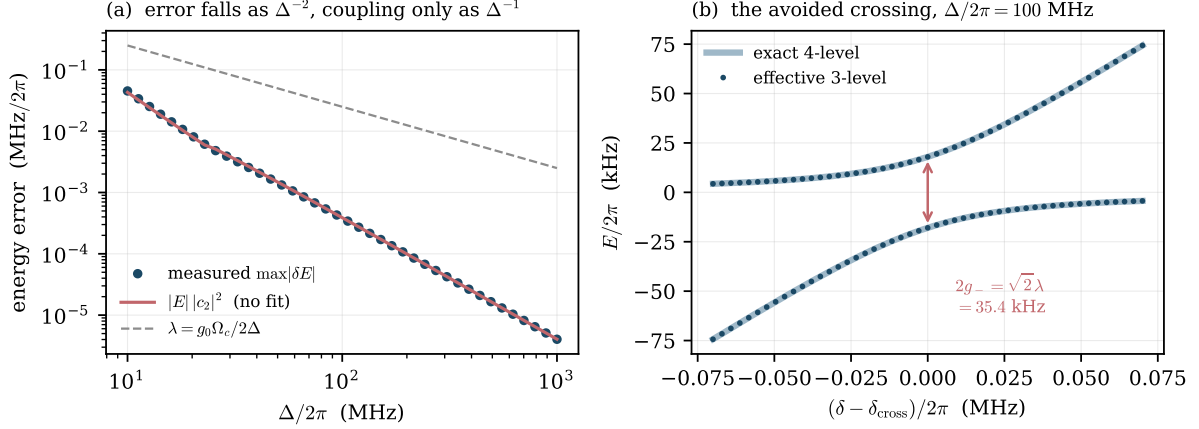


Figure 1: **Step 3 tested against exact diagonalisation** of Eq. (6) at $n = 1$, with $g_0/2\pi = 1$, $\Omega_c/2\pi = 5$, $\Omega_\mu/2\pi = 2$, $\delta/2\pi = 0.3$ MHz and $\Delta_\mu = 0$. (a) Worst-case eigenvalue error against detuning (points), compared with Eq. (21) (line), which contains no fitted quantity; the Raman coupling λ is drawn for scale, falling only as Δ^{-1} . (b) The avoided crossing between the cavity and the lower dressed branch at $\Delta/2\pi = 100$ MHz, whose width is the quantity Step 3 exists to produce. Measured minimum gap 35.9 kHz against the predicted $2g_- = \sqrt{2}\lambda = 35.4$ kHz; the 1.5% excess is the $\Omega_c^2/4\Delta$ light shift, which sits on $|3\rangle$ alone and therefore slightly reshapes the dressing of Step 4.

Where the dominant systematic comes from. The corrections in Eq. (12) are all built from the same three numbers V_m , so they cannot be adjusted independently: the Raman coupling $\lambda \propto \Omega_c$ and the light shift $\propto \Omega_c^2$ both track the control-laser intensity, and the shift adds *directly* to δ — the very quantity the measurand moves. Since $\Omega_c^2 \propto I$, a fractional intensity fluctuation ε displaces δ by $\varepsilon \Omega_c^2/4\Delta$, at the numbers above $\varepsilon \times 2\pi \times 62$ kHz. Control-laser intensity noise is therefore structurally indistinguishable from signal, and no amount of care with the microwave will separate them.

5 Step 4: dress the Rydberg pair

The lower-right block of Eq. (14) is a two-level problem on its own:

$$H_\mu = \begin{pmatrix} \delta & \Omega_\mu/2 \\ \Omega_\mu/2 & \delta + \Delta_\mu \end{pmatrix} = \left(\delta + \frac{\Delta_\mu}{2}\right) \mathbf{1} + \frac{1}{2}(\Omega_\mu \sigma_x + \Delta_\mu \sigma_z). \quad (22)$$

Diagonalise with the rotation $R(\theta)$ defined by

$$\tan \theta = \frac{\Omega_\mu}{\Delta_\mu}, \quad |+\rangle = \sin \frac{\theta}{2} |3\rangle + \cos \frac{\theta}{2} |4\rangle, \quad |-\rangle = \cos \frac{\theta}{2} |3\rangle - \sin \frac{\theta}{2} |4\rangle, \quad (23)$$

giving the *generalised Rabi* splitting

$$\omega_\pm = \delta + \frac{\Delta_\mu}{2} \pm \frac{W}{2}, \quad W = \sqrt{\Omega_\mu^2 + \Delta_\mu^2} \quad (24)$$

Now rotate the rest of Eq. (14). The Raman coupling connects $|1\rangle$ to the *bare* $|3\rangle = \sin \frac{\theta}{2} |+\rangle + \cos \frac{\theta}{2} |-\rangle$, so it splits between the branches in that ratio. In the basis $(|g\rangle, |+\rangle, |-\rangle)$:

$$H^{(3)} = \begin{pmatrix} 0 & g_+ a^\dagger & g_- a^\dagger \\ g_+ a & \omega_+ & 0 \\ g_- a & 0 & \omega_- \end{pmatrix}, \quad g_+ = \lambda \sin \frac{\theta}{2}, \quad g_- = \lambda \cos \frac{\theta}{2} \quad (25)$$

There is no $|+\rangle \leftrightarrow |-\rangle$ entry: that transition is the microwave one and carries no cavity matrix element.

Equation (25) is where this note stops. It is still excitation-number conserving, so on its own it radiates nothing; turning it into a Dicke model requires counter-rotating terms $a^\dagger b^\dagger + ab$ from a second drive tone, and that, together with the Holstein–Primakoff step and the superradiant threshold, is the subject of the companion note.

A The same elimination by hand

Section 4 eliminated $|2\rangle$ from the eigenvalue problem, which is the right thing to do when one wants energies. The same elimination can be carried out on the time-dependent problem, where the assumption being made — that $|2\rangle$ has no independent motion of its own — is easier to see. Expand the state as $|\Psi\rangle = \sum_j |j\rangle \otimes |\phi_j\rangle$ with $|\phi_j\rangle$ (unnormalised) cavity states. Reading the rows of Eq. (3),

$$i \partial_t |\phi_1\rangle = \omega_c a^\dagger a |\phi_1\rangle + g_0 a^\dagger |\phi_2\rangle, \quad (26)$$

$$i \partial_t |\phi_2\rangle = \omega_c a^\dagger a |\phi_2\rangle + \Delta |\phi_2\rangle + g_0 a |\phi_1\rangle + \frac{\Omega_c}{2} |\phi_3\rangle, \quad (27)$$

$$i \partial_t |\phi_3\rangle = \omega_c a^\dagger a |\phi_3\rangle + \delta |\phi_3\rangle + \frac{\Omega_c}{2} |\phi_2\rangle + \frac{\Omega_\mu}{2} |\phi_4\rangle, \quad (28)$$

and a fourth line for $|\phi_4\rangle$ that never involves $|\phi_2\rangle$. Only Eq. (27) carries a large coefficient. Remove the slow diagonal terms with $|\phi_j\rangle = e^{-i\omega_c a^\dagger a t} e^{-i\epsilon_j t} |\chi_j\rangle$, $\epsilon_{1,3,4} = 0, \delta, \delta + \Delta_\mu$, and it becomes

$$i \partial_t |\chi_2\rangle = \Delta |\chi_2\rangle + \underbrace{g_0 a e^{-i\omega_c t} |\chi_1\rangle + \frac{\Omega_c}{2} e^{-i\delta t} |\chi_3\rangle}_{S(t)}, \quad (29)$$

a driven first-order equation whose source has amplitudes $g_0 \sqrt{\langle n \rangle}$, $\Omega_c/2$ and turns over at rates ω_c, δ — condition (A2), item by item. When $|\Delta|$ dominates all of them, $|\chi_2\rangle$ never builds up: it locks to the value that makes the right-hand side vanish, $|\chi_2\rangle \simeq -S(t)/\Delta$, i.e.

$$|\phi_2\rangle \simeq -\frac{1}{\Delta} \left(g_0 a |\phi_1\rangle + \frac{\Omega_c}{2} |\phi_3\rangle \right). \quad (30)$$

$|2\rangle$ is no longer a degree of freedom; it is a shadow of the two states that drive it. Substituting into Eq. (26),

$$g_0 a^\dagger |\phi_2\rangle = -\frac{g_0^2}{\Delta} a^\dagger a |\phi_1\rangle - \frac{g_0 \Omega_c}{2\Delta} a^\dagger |\phi_3\rangle, \quad (31)$$

and into Eq. (28),

$$\frac{\Omega_c}{2} |\phi_2\rangle = -\frac{g_0 \Omega_c}{2\Delta} a |\phi_1\rangle - \frac{\Omega_c^2}{4\Delta} |\phi_3\rangle. \quad (32)$$

These are the same four paths tabulated in §4.4, and they assemble into the same Eq. (14). The correspondence with §4.2 is exact: Eq. (30) is Eq. (10) with E already dropped from the denominator, and substituting it back into the remaining rows multiplies it by V_m , which is Eq. (12). The two calculations answer different questions — one the stationary energies, the other the motion of an arbitrary state — and make the same single approximation.